

Computer Vision

Introduction of Object Detection by
YOLO Algorithm

YOLO

You Only Look Once

Yolo is used for Object Detection

Detection = Classification + Localization

Classification



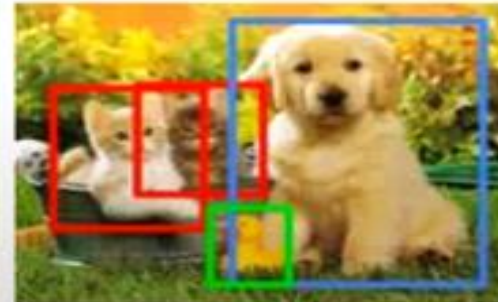
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

How YOLO works?

Our task is to predict a class of an object and the bounding box specifying object location.

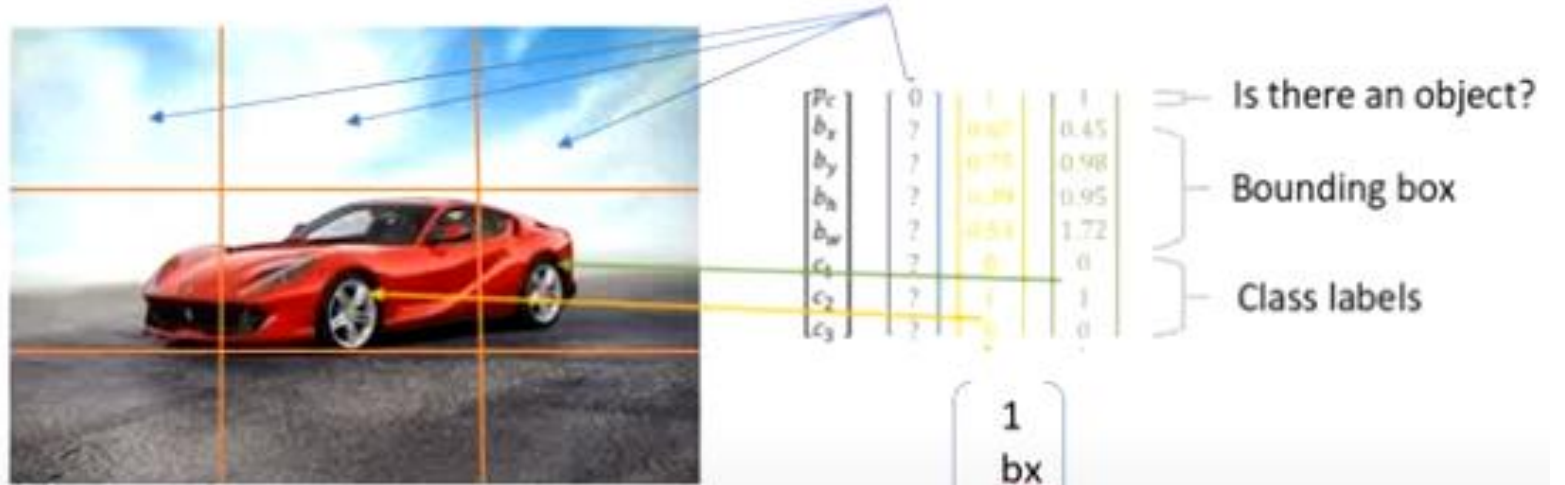
Each bounding box can be described using four descriptors:

1. center of a bounding box (b_x, b_y)
2. width (b_w)
3. height (b_h)
4. c class of an object (Eg car, truck, phone etc)



But lets Understand the concept with 3*3 grid cell

Suppose we want to detect
3 objects –Truck, Car, Cycle



$$Y = \begin{pmatrix} 0 \\ ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{pmatrix}$$

$$Y = \begin{pmatrix} 1 \\ \underline{b_x} \\ \underline{b_y} \\ \underline{b_h} \\ \underline{b_w} \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

Intersection over union method

Intersection over union (IoU) computes the intersection over union on these 2 bounding blocks

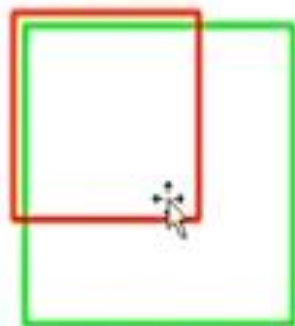


$$\text{intersection over union}_I = \frac{\text{Size of intersection}}{\text{Size of union}}$$

"Correct" if $\text{IoU} \geq 0.5$

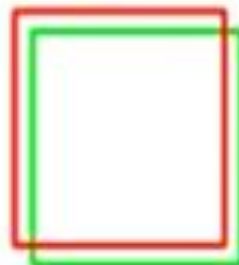
IoU

IoU: 0.4034



Poor

IoU: 0.7330



Good

IoU: 0.9264

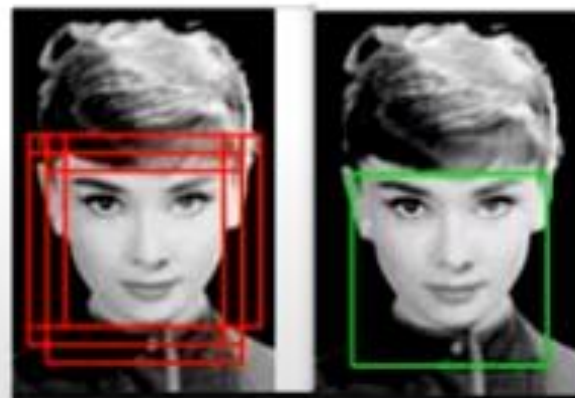


Excellent

NON-MAX SUPPRESSION

Non-max suppression is a common algorithm used for cleaning up when multiple boxes are predicted for the same object.

We have used 3 x 3 bounding boxes.



Non-max suppression

Non-max suppression clean up our results:

1. Discard all boxes with pc less or equal to 0.6.
2. Pick the box with the largest pc output as a prediction.
3. Discard any remaining box with IoU greater than or equal to 0.5.

And finally, the cleaned up prediction looks like this:



Anchor Boxes

Anchor boxes are a set of predefined bounding boxes of a certain height and width.

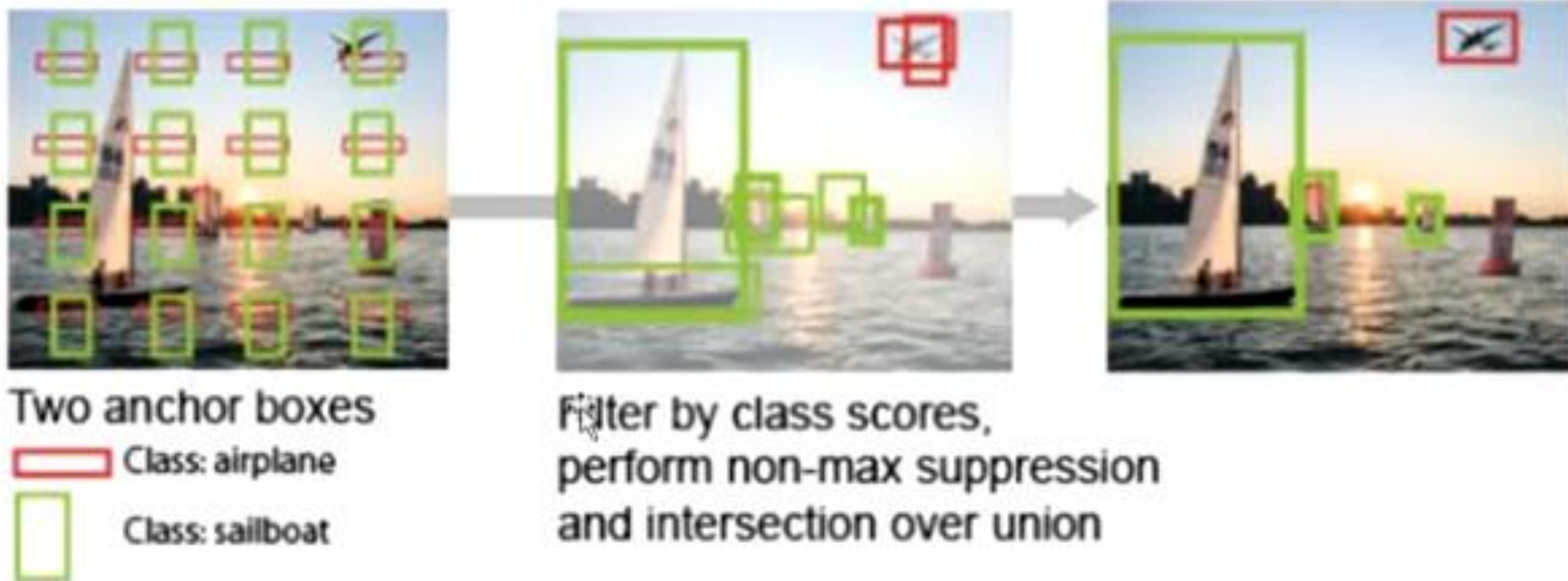


Image Classification

Is this a dog or a person?



Neural
Network
Output

Dog = 1
Person = 0

Object Localization

Where exactly is the dog in
this image?



Neural
Network
Output

Dog = 1
Person = 0
+
Bounding
Box

Object Localization



P_c	1
B_x	50
B_y	70
B_w	60
B_h	70
C_1	1
C_2	0

$C_1 = \text{Dog class}$
 $C_2 = \text{Person Class}$



1
30
28
28
82
0
1



0
-
-
-
-
-
-

X_{train}



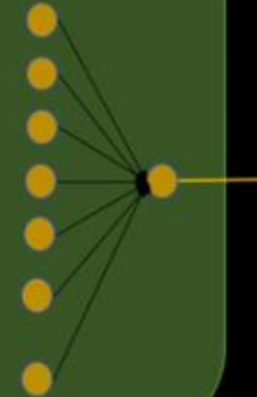
y_{train}

p_c	1
B_x	50
B_y	70
B_w	60
B_h	70
C_1	1
C_2	0

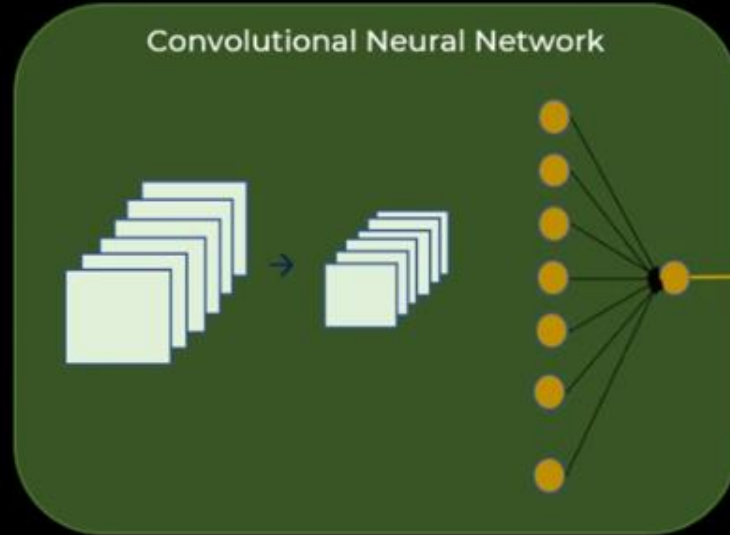
1
30
55
28
82
0
1

0
-
-
-
-
-
-

Convolutional Neural Network



p_c
B_x
B_y
B_w
B_h
C_1
C_2

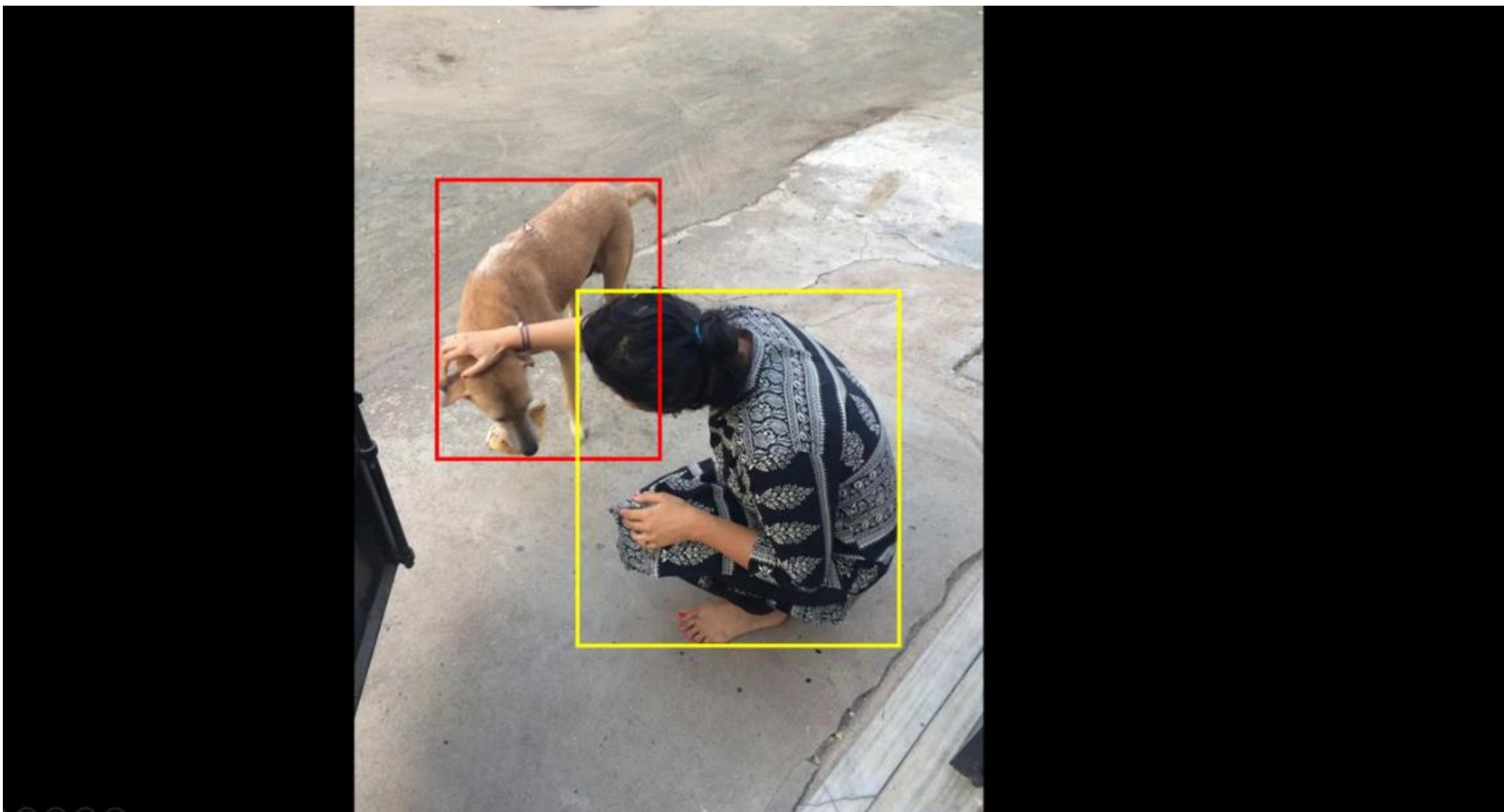


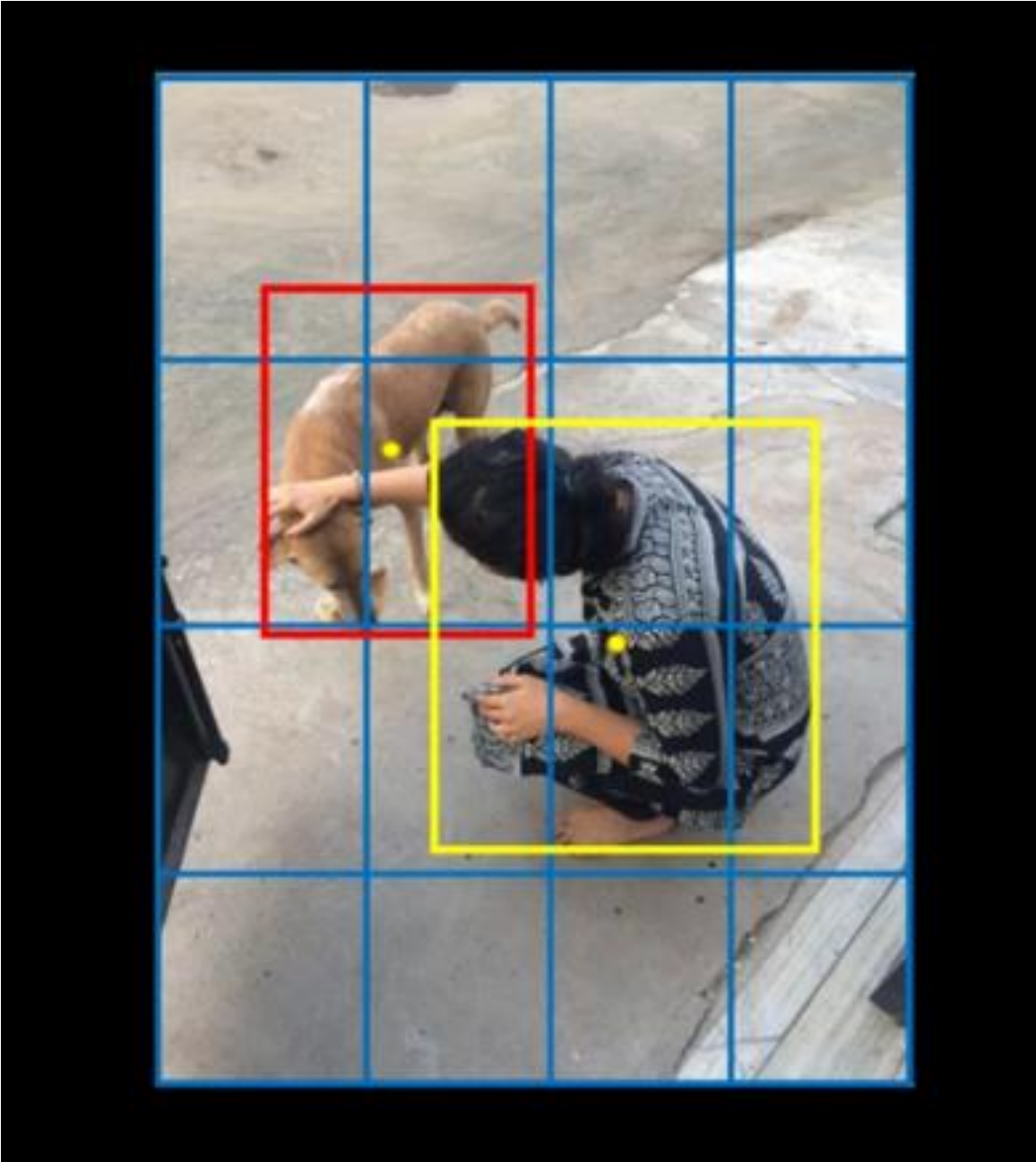
1
25
57
30
42
1
0



This works ok
only for single
object. What
about multiple
objects in an
image?

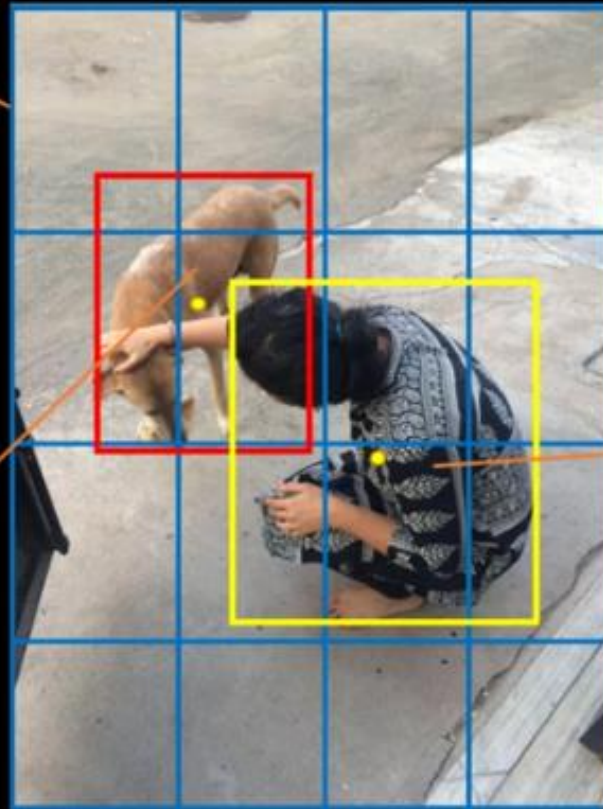
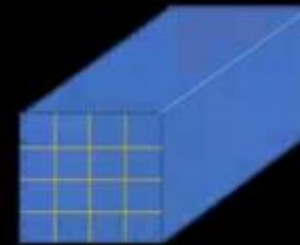






4 by 4 by 7

$$\begin{bmatrix} P_c \\ B_x \\ B_y \\ B_w \\ B_h \\ C_1 \\ C_2 \end{bmatrix} \begin{bmatrix} 0 \\ - \\ - \\ - \\ - \\ - \\ - \end{bmatrix}$$



$$\begin{bmatrix} 1 \\ 0.05 \\ 0.3 \\ 2 \\ 1.3 \\ 1 \\ 0 \end{bmatrix}$$

(0,0)



(1,1)

$$\begin{bmatrix} 1 \\ 0.32 \\ 0.02 \\ 3 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$



X_{train}



y_{train}

16 such
vectors



$\begin{bmatrix} p_c \\ B_x \\ B_y \\ B_w \\ B_h \\ C_1 \\ C_2 \end{bmatrix}$

16 such
vectors



$\begin{bmatrix} p_c \\ B_x \\ B_y \\ B_w \\ B_h \\ C_1 \\ C_2 \end{bmatrix}$

16 such
vectors



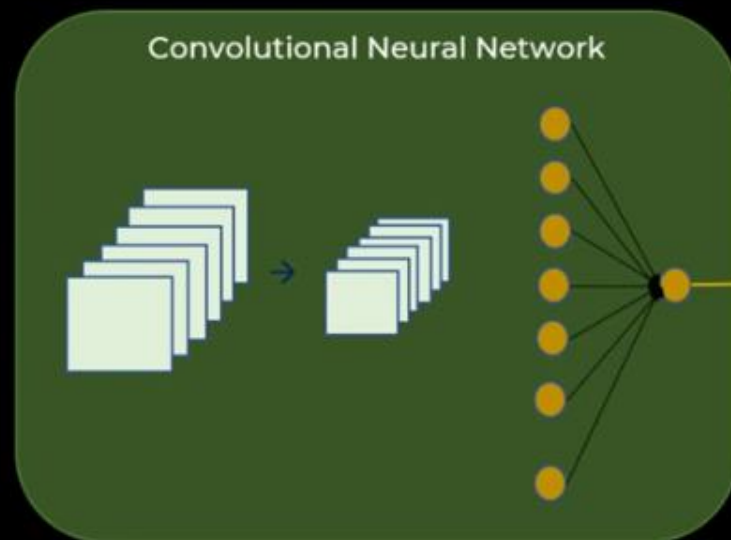
$\begin{bmatrix} p_c \\ B_x \\ B_y \\ B_w \\ B_h \\ C_1 \\ C_2 \end{bmatrix}$



Convolutional Neural Network



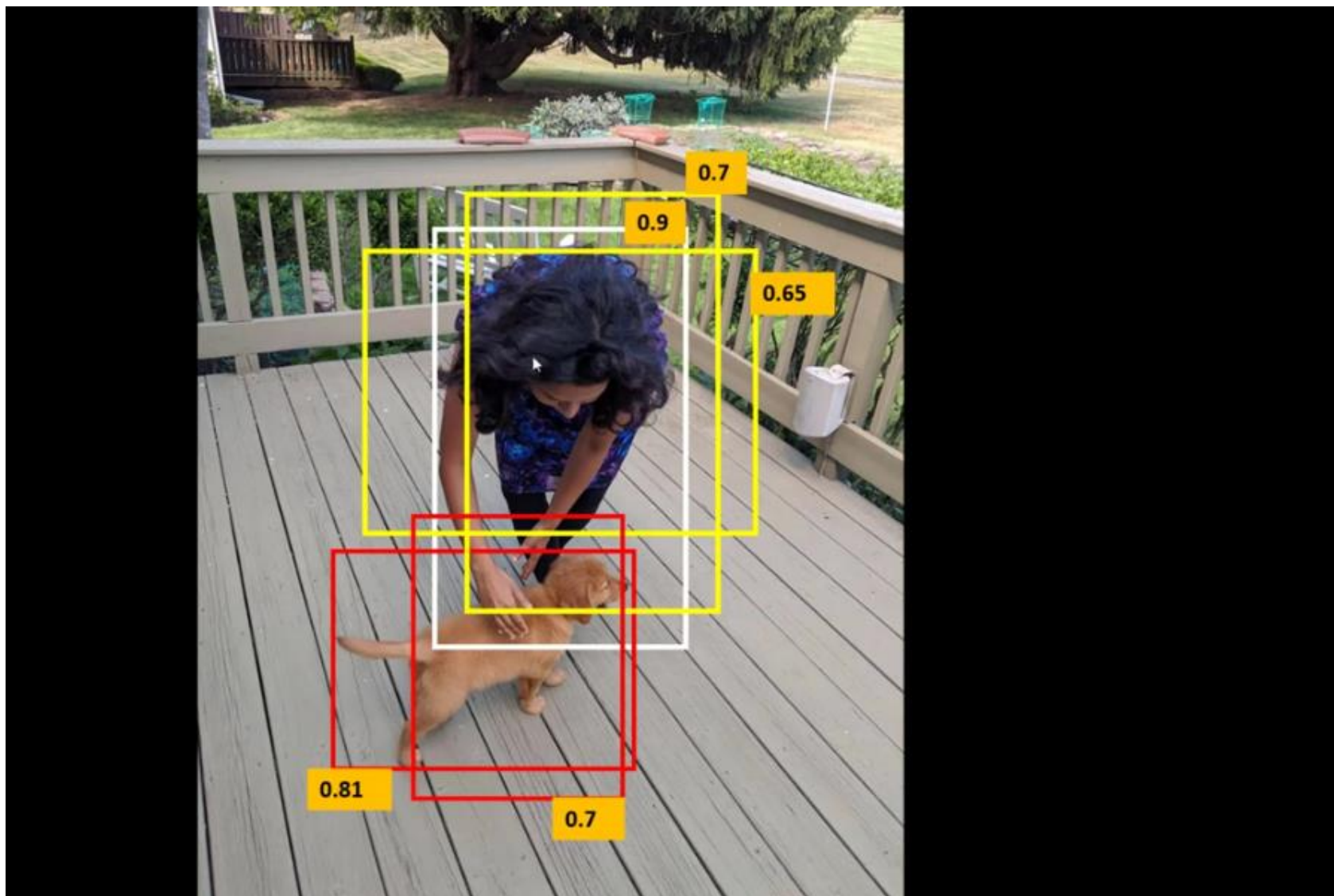
Prediction

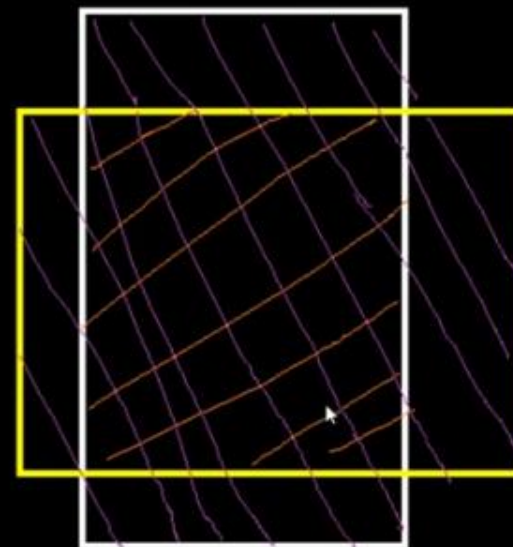
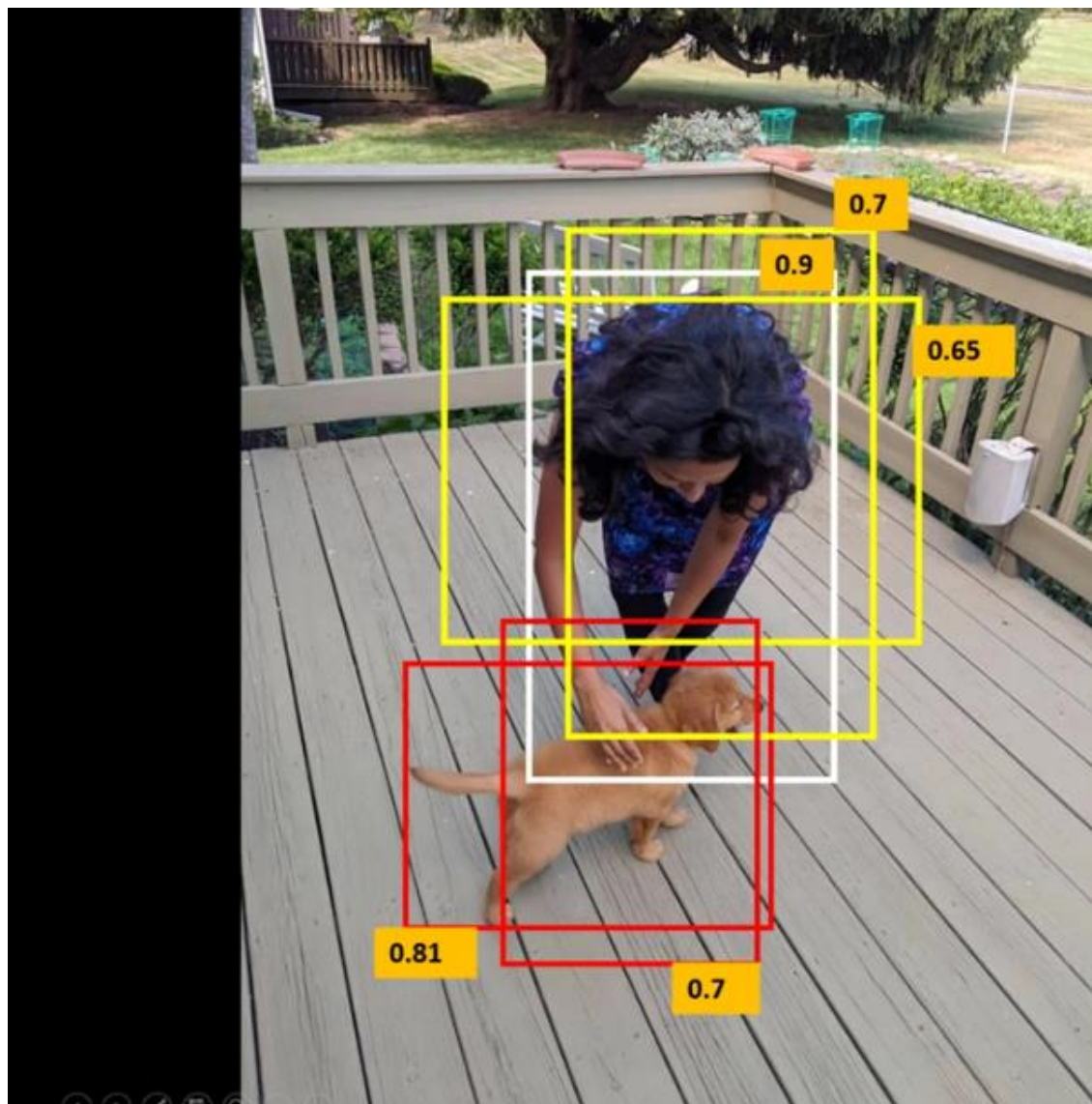


16 such vectors

$\begin{bmatrix} p_c \\ B_x \\ B_y \\ B_w \\ B_h \\ C_1 \\ C_2 \end{bmatrix}$

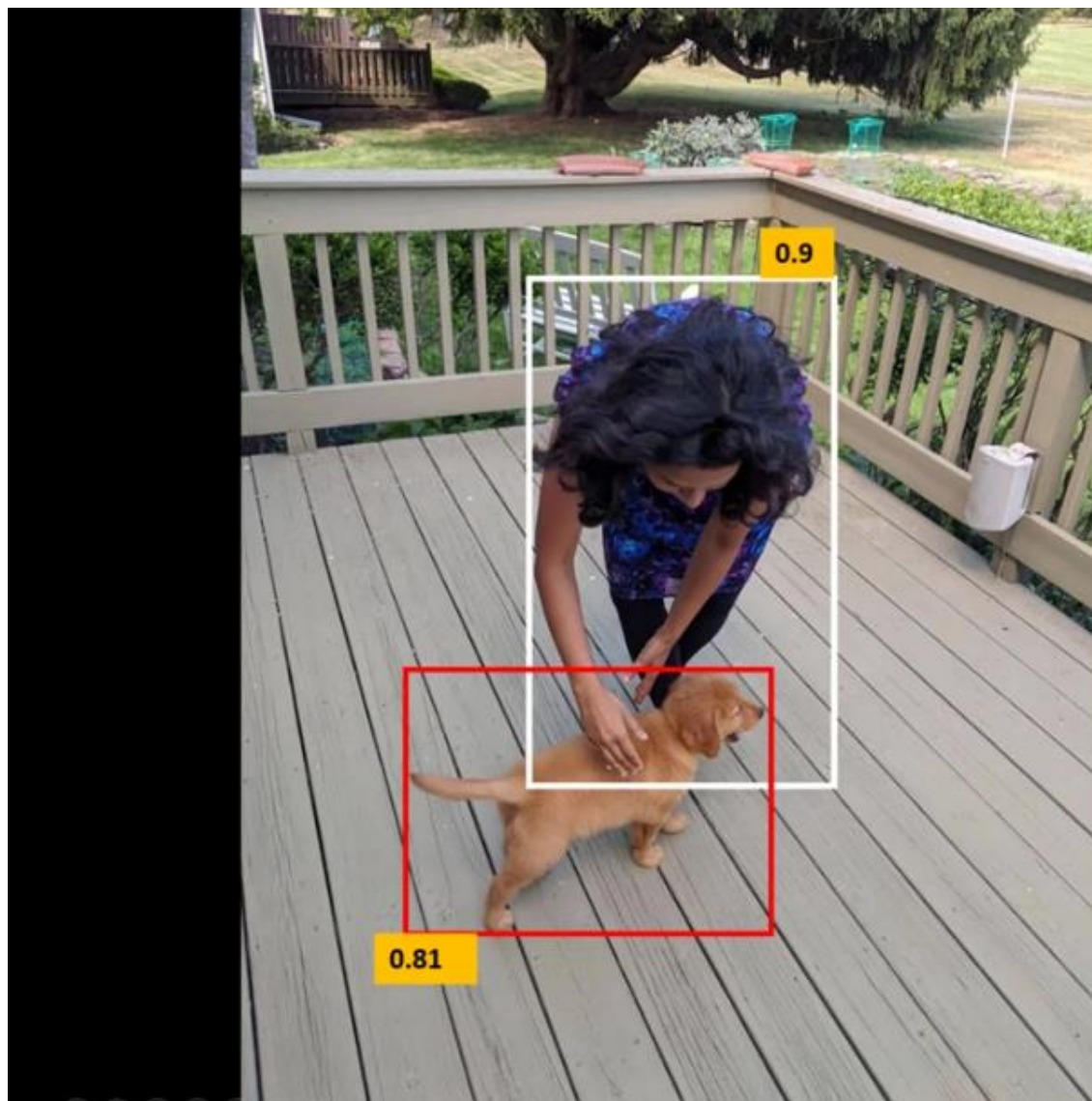






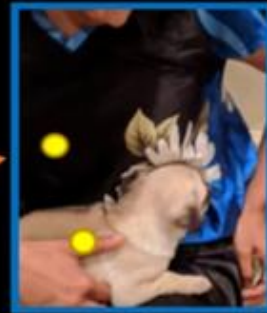
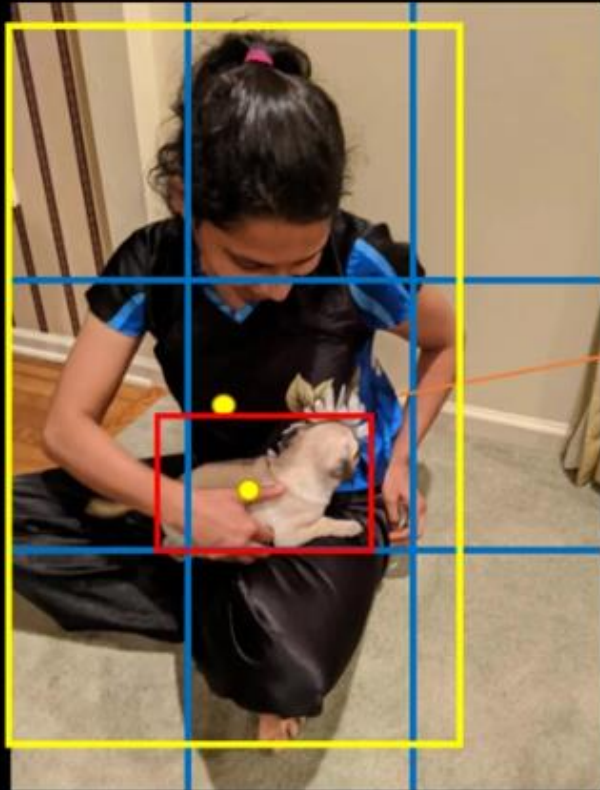
Intersection over union = $\text{intersect area} / \text{union area}$

Intersection over union : IOU



Non max
suppression

What if one grid cell has center of two objects?



P_c
 B_x
 B_y
 B_w
 B_h
 C_1
 C_2

1
 0.22
 0.45
 1
 0.7
 1
 0

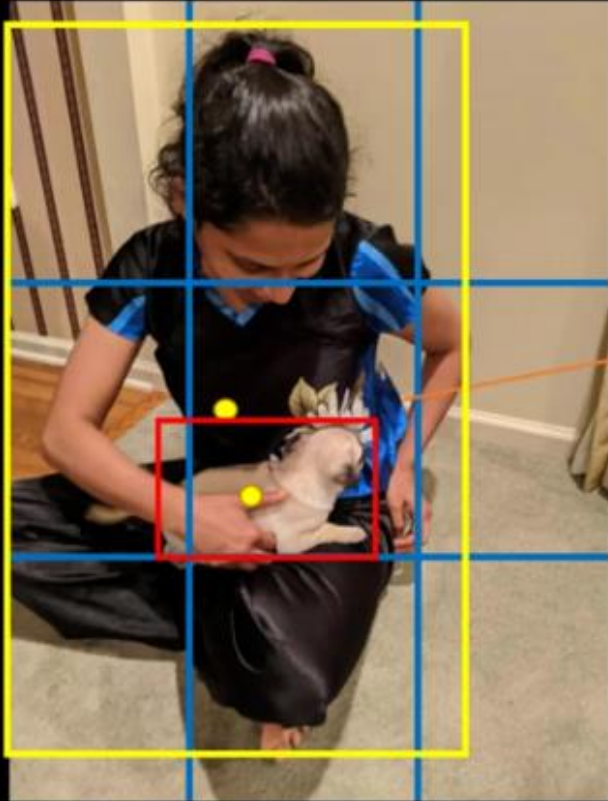
Dog

1
 0.13
 0.38
 2.1
 3
 0
 1

Person

1
 0.22
 0.45
 1
 0.7
 1
 0
 1
 0.13
 0.38
 2.1
 3
 0
 1

This concept is called anchor boxes

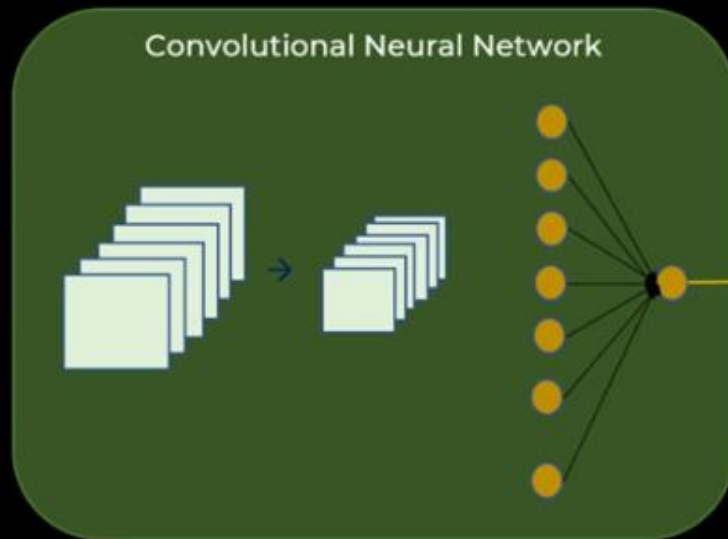


1
0.22
0.45
1
0.7
1
0
1
0.13
0.38
2.1
3
0
1

Two anchor
boxes

You can have
more

CNN with Two anchor boxes



16 such vectors

P_{c1}
 B_{x1}
 B_{y1}
 B_{w1}
 B_{h1}
 C_{11}
 C_{21}
 P_{c2}
 B_{x2}
 B_{y2}
 B_{w2}
 B_{h2}
 C_{12}
 C_{22}

